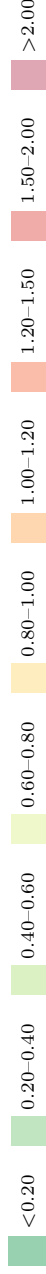


Task type	Qwentaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]
	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	Centaur-70B
Decision (2)	0.30	0.32	0.35	0.34	0.37	0.27	0.31	0.33	0.20	0.21	0.24	0.34	0.21	0.31	0.28	0.29	0.28	0.33
MDP (2)	0.56	0.56	0.58	<u>0.58</u>	<u>0.58</u>	0.48	0.53	0.57	0.39	0.46	0.52	0.57	0.42	0.57	0.51	0.53	0.53	0.50
Bandit (4)	0.32	0.35	0.37	0.36	<u>0.38</u>	0.33	0.33	0.36	0.24	0.23	0.32	0.34	0.28	0.34	0.22	0.25	0.25	0.39
Misc. (7)	0.28	0.31	0.45	0.46	<u>0.49</u>	0.14	0.30	0.40	0.03	0.11	0.18	0.36	0.09	0.35	0.37	0.41	0.29	0.52
Mean (15)	0.33	0.35	0.43	0.43	<u>0.45</u>	0.26	0.34	0.40	0.16	0.20	0.27	0.38	0.20	0.37	0.34	0.37	0.31	<u>0.46</u>
<0.05 0.05–0.12 0.12–0.18 0.18–0.25 0.25–0.32 0.32–0.40 0.40–0.48 0.48–0.55 ≥0.55																		

Table 1: Fraction of available information captured above chance, $(\ln k - \text{NLL}) / \ln k$, by task type on Psych-201 (out-of-distribution). A value of 0 indicates chance-level performance; 1 indicates perfect prediction. Restricted to 15 experiments (of 18) with a well-defined discrete response space ($\ln k > 0$); 3 experiments with continuous or mixed responses are excluded. **Bold+underline** marks the best model, underline the second-best.

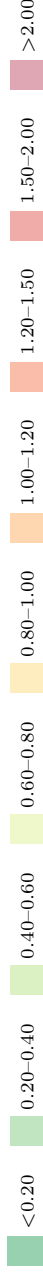
Experiment	Type	Queen3 family												Llama-3.1/3.2 family												[binz2025foundation]				Chance
		0.6B			1.7B			4B			8B			14B			1B			3B			7B			ln(k)				
		base	ft		base	ft		base	ft		base	ft		base	ft		base	ft		base	ft		base	ft						
Experience-description choice [anilo2024weird]	Decision	0.50	0.42		0.49	0.40		0.46	0.39		0.46	0.38	0.45	0.38	0.49	0.44		0.46	0.41		0.46	0.41		0.46	0.41		0.69			
Risky decision and happiness [ruthledge2023happiness]	Decision	1.46	1.38		1.45	1.34		1.39	1.31	1.33	1.57	1.40	1.53	1.37	1.43	1.34		1.53	1.32		1.51			1.32			0.69			
Loss aversion task [spektor2024lossaversion]	Decision	0.57	0.54		0.56	0.54		0.54	0.52		0.53	0.54	0.53	0.50	0.66	0.57		0.61	0.55		0.54	0.52		0.54	0.52		0.69			
Two-step task [castrorodriguez2022twostep]	MDP	0.26	0.26		0.27	0.28		0.26	0.24		0.26	0.24	0.25	0.25	1.29	0.45		0.95	0.35		0.26	0.25		0.46			1.39			
Two-step task [shahar2019twosteptask]	MDP	0.55	0.48		0.53	0.47		0.54	0.47		0.55	0.47	0.53	0.46	0.54	0.49		0.53	0.47		0.53	0.47		0.46			0.69			
Range adaptation RL [bavard2021range]	Bandit	0.56	0.48		0.54	0.48		0.53	0.47		0.52	0.47	0.52	0.47	0.57	0.48		0.52	0.48		0.52	0.46		0.45			0.69			
Explore-exploit task [fan2022trait]	Bandit	0.42	0.28		0.40	0.27		0.40	0.26		0.39	0.27	0.40	0.26	0.42	0.28		0.42	0.28		0.42	0.26		0.26			0.69			
Counterfactual learning [palminteri2017confirmation]	Bandit	0.64	0.50		0.62	0.49		0.60	0.48		0.61	0.49	0.58	0.47	0.62	0.53		0.58	0.49		0.58	0.48		0.46			0.69			
Contextual reinforcement learning [vandendriessche2022depression]	Bandit	0.69	0.61		0.69	0.57		0.61	0.54		0.63	0.55	0.57	0.53	0.69	0.58		0.59	0.61		0.59	0.56		0.51			0.69			
Aversive reversal learning [zika2023traitanxiety]	Bandit	3.66	3.36		3.70	3.36		3.57	3.29		3.51	3.27	3.47	3.27	4.00	3.99		3.90	3.85		3.87	3.82		3.80			—			
Navon task [busch2024navon]	Misc.	0.43	0.40		0.40	0.35		0.43	0.32	0.41	0.32	0.37	0.32	0.47	0.59	0.44		0.44	0.45		0.39	0.34		0.31			0.69			
Stroop task [busch2024stroop]	Misc.	0.53	0.38		0.28	0.63		0.21	0.18		0.20	0.17	0.21	0.15	0.97	0.85		0.29	0.45		0.20	0.18		0.15			1.10			
Pragmatic language task [franke2024bayesian]	Misc.	1.55	1.25		1.47	1.03		1.17	0.81		1.08	0.76	1.07	0.77	1.64	1.49		1.75	1.25		1.85	1.07		0.75			1.39			
Reference game [frankeden2016reasoning]	Misc.	1.50	1.21		1.33	1.12		1.04	0.83		1.14	0.85	0.97	0.80	1.47	1.34		1.41	1.12		1.19	1.04		0.80			1.39			
Compound word processing [guenther2020ts]	Misc.	0.62	0.49		0.52	0.41		0.37	0.37		0.36	0.37	0.34	0.33	0.49	0.47		0.42	0.38		0.37	0.34		0.32			0.69			
Grammaticality judgement [guenther2023grammaticality]	Misc.	0.47	0.43		0.37	0.41		0.35	0.36		0.35	0.31	0.32	0.30	0.50	0.42		0.40	0.39		0.34	0.33		0.29			0.69			
Scalar inference [tsvilodub2023xorsome]	Misc.	3.37	3.33		3.29	3.19		3.13	3.04		3.06	3.00	2.97	2.95	3.62	3.50		3.48	3.38		3.23	3.20		2.91			—			
Math arithmetic task [xu2023augmenting]	Misc.	0.72	0.72		0.71	0.71		0.71	0.70		0.69	0.70	0.68	0.66	0.73	0.72		0.73	0.72		0.71	0.72		0.55			0.69			
Mean (18)		1.03	0.92		0.98	0.89		0.91	0.81		0.90	0.80	0.87	0.79	1.15	1.03		1.06	0.94		0.97	0.88		0.82			0.86			



Bold+underline = best; underline = second-best. Lower is better. “—” = continuous or mixed response space. Size labels in billions of parameters.

Table 2: Per-experiment NLL on Psych-201 (out-of-distribution). Psych-201 was not used during supervised fine-tuning; lower NLL indicates better generalisation. Within each model size, the left column (base) shows the pretrained model and the right column (ft) shows the cognitively finetuned variant. The $\ln(k)$ column shows the random-guessing baseline where k is the number of per-trial response options.

Experiment	Type	Qentaur				Llama-Centaur				Smol'taur				Olmotaur				Chance	
		0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	70B	ln(k)		
Experience-description choice [anllo2024weird]	Decision	0.42	0.40	0.39	0.38	0.38	0.44	0.41	0.41	0.53	0.52	0.47	0.40	0.49	0.41	0.42	0.69		
Risky decision and happiness [rutledge2023happiness]	Decision	1.38	1.40	1.34	1.31	1.33	1.40	1.37	1.34	1.50	1.44	1.46	1.36	1.48	1.38	1.32	—		
Loss aversion task [spektor2024lossaversion]	Decision	0.54	0.54	0.52	0.54	0.50	0.57	0.55	0.52	0.58	0.58	0.58	0.52	0.61	0.54	0.51	0.69		
Two-step task [castrorodriguez2022twostep]	MDP	0.26	0.28	0.24	0.24	0.25	0.45	0.35	0.25	0.48	0.43	0.34	0.25	0.47	0.26	0.46	1.39		
Two-step task [shahar2019twosteptask]	MDP	0.48	0.47	0.47	0.47	0.46	0.49	0.47	0.47	0.61	0.54	0.50	0.47	0.58	0.47	0.46	0.69		
Range adaptation RL [bavard2021range]	Bandit	0.48	0.48	0.47	0.47	0.47	0.48	0.48	0.46	0.54	0.55	0.51	0.48	0.55	0.48	0.45	0.69		
Explore-exploit task [fan2022trait]	Bandit	0.28	0.27	0.26	0.27	0.26	0.28	0.27	0.26	0.35	0.32	0.29	0.27	0.29	0.27	0.26	0.69		
Counterfactual learning [palminteri2017confirmation]	Bandit	0.50	0.49	0.48	0.49	0.47	0.53	0.49	0.48	0.59	0.60	0.52	0.52	0.59	0.51	0.46	0.69		
Contextual reinforcement learning [vandendriessche2022depression]	Bandit	0.61	0.57	0.54	0.55	0.53	0.58	0.61	0.56	0.64	0.68	0.58	0.57	0.57	0.58	0.51	0.69		
Aversive reversal learning [zika2023traitanxiety]	Bandit	3.36	3.36	3.29	3.27	3.27	3.99	3.85	3.82	3.54	3.44	3.33	3.89	4.35	4.00	3.80	—		
Navon task [busch2024navon]	Misc.	0.40	0.35	0.32	0.32	0.32	0.59	0.45	0.34	0.65	0.61	0.44	0.34	0.72	0.35	0.31	0.69		
Stroop task [busch2024stroop]	Misc.	0.38	0.63	0.18	0.17	0.15	0.85	0.45	0.18	0.67	0.76	0.68	0.29	0.88	0.39	0.15	1.10		
Pragmatic language task [franke2024bayesian]	Misc.	1.25	1.03	0.81	0.76	0.77	1.49	1.25	1.07	1.61	1.64	1.54	1.06	1.39	1.08	0.75	1.39		
Reference game [frankedegen2016reasoning]	Misc.	1.21	1.12	0.83	0.85	0.80	1.34	1.12	1.04	1.92	1.35	1.45	1.14	1.38	1.04	0.80	1.39		
Compound word processing [guenther2020ts]	Misc.	0.49	0.41	0.37	0.37	0.33	0.47	0.38	0.34	0.61	0.52	0.44	0.35	0.56	0.42	0.32	0.69		
Grammaticality judgement [guenther2023grammaticality]	Misc.	0.43	0.41	0.36	0.31	0.30	0.42	0.39	0.33	0.49	0.45	0.42	0.40	0.45	0.35	0.29	0.69		
Scalar inference [tsvilodub2023xorsome]	Misc.	3.33	3.19	3.04	3.00	2.95	3.50	3.38	3.20	3.78	3.60	3.45	3.28	3.96	3.30	2.91	—		
Math arithmetic task [xu2023augmenting]	Misc.	0.72	0.71	0.70	0.70	0.66	0.72	0.72	0.72	0.78	0.75	0.74	0.72	0.75	0.72	0.55	0.69		
Mean (18)		0.92	0.89	0.81	0.80	0.79	1.03	0.94	0.88	1.10	1.04	0.99	0.91	1.11	0.92	0.82	0.86		



Bold+underline = best; underline = second-best (across the four families). Lower is better. "—" = continuous or mixed response space. Size labels in billions of parameters. 70B is the reproduced Centaur-70B (reference only; excluded from marking).

Table 3: Per-experiment NLL on Psych-201 (out-of-distribution) for the four finetuned families. Psych-201 was not used during fine-tuning; lower NLL indicates better generalisation. The $\ln(k)$ column shows the random-guessing baseline where k is the number of per-trial response options.